

Final Exam

Stat 158, Spring 2025

Name: _____

SID: _____

Instructions

1. This is a pencil/pen and paper exam. No electronic devices are needed or permitted. Any mathematical work can be left unsimplified.
2. The only outside source that you may consult is your own handwritten optional cheat-sheet.
3. Write your answers on this exam on the same page as the question.
4. If you need scratch paper, please use the final sheet of this exam.
5. Please write neatly.
6. Good luck!

Identify the Design

For each of the following situations, give the response, factors, units, and type of design. For each factor, indicate whether it is experimental/observation/blocking and the number of levels. Describe the potential outcomes and how many there are for each unit.

1. Researchers want to study how phone notification settings affect students' focus during study sessions. One setting they consider is frequency (high vs. low); the second is tone (sound on vs. vibrate). 24 students are randomly assigned to a level of both frequency and tone, with an equal number of students in each group. Each student studies for 45 minutes while their phone receives simulated messages. The outcome is the number of correctly answered questions on a quiz administered immediately afterwards.
2. A public policy researcher is analyzing how commuting method and income level are associated with well-being in a sample of 600 Bay Area residents. Participants are classified by commuting method (car, public transit, bicycle, or walk) and income category (low, medium, high). The outcome is their score on a 10-item subjective well-being scale. The researcher uses existing survey data.
3. Public health researchers draft a script of helpful questions for doctors to ask patients during interactions in the clinic. Six of the 12 doctors practicing in a single facility are chosen at random to receive copies of the script and a brief training about how to use it. The next day, each of the 12 doctors sees 8 patients and treats them all according to the script. The public health researchers interview each of the 96 patients immediately after their visit and have each patient rate their satisfaction with the care they received on a scale from 1 to 5.

Short Answer

4. The final study described in the previous section of this exam involves a response measured on a scale from 1 to 5. Which inferential framework is more appropriate here: randomization-based or model-based (using Gaussian errors with constant variance)? Justify your response and reference the particularities of this study.

5. What is the difference between an *experimental unit* and a *measurement unit*? Answer using their definitions and then formulate a (brief) novel experimental setting that demonstrates the distinction.

6. What is *interference* in an experimental setting? Answer using the definition and then formulate a (brief) novel experimental setting that demonstrates this phenomenon.

Study 1: Burn Treatment

During WWII in England, men volunteered to be part of research to investigate how to best treat burns from possible chemical gas attacks from Germany. The volunteers agreed to be given small burns with mustard gas to assess what treatment of victims would be most beneficial. The statistical analysis of this work was overseen by a researcher, George E. P. Box (1919-2013), who was to become a preeminent statistician in the area of experimental design.

Six burn medications (including the control of no medication) were tested. Burns heal differently on different parts of the body, so six sites on the arms were considered. After each medication was applied, the burn was covered by a clean gauze; the response was the number of hours for a clearly defined degree of partial healing to occur.

Consider the following completely randomized design: the combination of burn medications (6 levels) and locations (6 levels) yields 36 treatment combinations. Each volunteer will be randomly assigned one of these 36 treatment combinations: a burn will be created on the volunteer at the site assigned and then the assigned treatment will be applied to the burn.

7. How could you make the design a complete block design? Describe your blocking factor(s), what your units would be, and how you would use randomization to assign treatments to units. What are advantages and drawbacks of such a design compared to the CR[2]?

8. How could you make the design a Latin Square? Describe your blocking factor(s), what your units would be, and how you would use randomization to assign treatments to units. What are advantages and drawbacks of such a design compared to the CR[2] or your complete block?

Study 2: Washing Bacteria

Two different washing solutions are compared to study their effectiveness in retarding bacteria growth. The washing solutions are applied to randomly selected five-gallon milk jugs and the response are the levels of bacteria remaining in the jugs after a half of a day. Lower values of the response indicated a more effective washing solution. Four milk jugs can be studied each day. Because ambient temperature is known to affect bacterial growth, the researchers decide to block on day to account for daily temperature differences.

Here is the data from the experiment (left table) and the group means (right table):

day	solution	bacteria		
1	1	13		
1	2	15		
1	1	22	solution	avg bacteria
1	2	24	1	23
2	1	18	2	25
2	2	17		
2	1	39		
2	2	44		

- You wish to construct a randomization-based confidence interval for the difference in effectiveness between Solutions 1 and 2. For this question only, assume the treatment effect of solution is constant across milk jugs, that there is no block effect, and that no interference is present. Create a full schedule of potential outcomes giving the real or imputed values for each of the potential outcomes for each unit in the study.

10. Next, you pursue a randomization-based test for the average treatment effect of the solution by reshuffling the original data set according to the block design. Write out a data set that you might have obtained by this reshuffling (choose one that is different from the original data).

[illegible]

11. Provide a numbered list of the steps to execute (six or fewer) in order calculate a **two-sided** p-value.

The following questions relate to the design phase of this experiment, conducted before collecting data.

12. Define *statistical power*.
13. Use model-based inference to find the power associated with a **two-sided** test of the difference in mean bacteria level between the two solutions when the true difference, $\mu_1 - \mu_2$, is 1, the residual variance in the treatment groups, σ^2 , is 4, and the number of units, N , is 8. No need to fully calculate for the power but set up the power calculation as a function of the appropriate PDFs/CDFs.
14. If this experiment used a completely randomized instead of a block design, what component of the power calculation would change? What would the consequence be for the power of the test?

Study 3: The Low Carb Diet

A researcher enrolls 3,000 subjects in a study and randomly assigns 1,000 of them to eat a low carbohydrate diet for a month. The remaining 2,000 are given no dietary guidance (the control). At the end of the month, researchers measure whether each subject can run a mile in a prespecified time. When subjects are tested at the end of the month, 500 out of the 1,000 subjects in the treated group report that they ate the diet; however, in reality only 250 of them did so. The results of the experiment show that 400 out of 1,000 subjects assigned to treatment ran the mile within the prespecified time period, and that 700 out of the 2,000 subjects assigned to control did so.

15. Explain why it is not possible to estimate well the ATE from this study.

16. If you believed that 500 subjects actually ate the diet, what would your estimate of the CACE be?

17. If instead you discovered that only 250 subjects had eaten the diet, what would your new estimate of the CACE be?

18. Do the subjects' exaggerated claims about their compliance with treatment make the diet seem more or less effective? You may define effectiveness with respect to either the CACE or the ITT_Y .

19. CACE is a parameter of which subset of subjects? How are they defined?

Study 4: Teaching AI to Teach

Researchers at UC Berkeley are studying whether AI chatbots can be tuned to increase students' understanding of an academic topic. They plan to recruit and compensate 40 UC Berkeley students to participate in a one-hour lesson. In the first 15 minutes each student learns about a new statistical topic either by reading about it or by watching a live lecturer. That is followed by 25 minutes where they work on a handout with practice problems, during which they are encouraged to use one of two AI chatbots. In the final 20 minutes they are given a quiz that they take without the chatbot. Their quiz score out of 100 serves as the measure of the student's understanding.

One of the chatbots is an answerer-AI that behaves like ChatGPT: if you ask it a question, it will provide a complete answer. The other chatbot is a teacher-AI that, when asked a question, behaves more similarly to a human teacher: it doesn't immediately provide an answer but asks the student to explain what they know and asks follow-up questions.

Each student will be randomly assigned to either be taught via reading or lecture; and to interact either with a teacher-AI or an answerer-AI; with an equal number of students in each group.

20. Identify the components of the experiment: the response, the factors and whether they are experimental / blocking / observational, the levels of each factor, and the experimental unit.

21. What type of design is used in this study?

22. How many rows and columns does the full schedule of potential outcomes have?

23. Write an algebraic model for the data from this experiment that includes main effects and interactions. Define each parameter, and clearly specify the range of values for any indices (e.g., how many levels for each factor).

Once the researchers began to write their protocol, they realized there was a constraint: they didn't have the time or space to study each student individually. Instead, they decided to direct students to sign up for one of 4 sessions, each taking place in a classroom with 10 total students. Each classroom will be randomly assigned to receive the lesson either via live lecture or through printed readings. As before, each individual student in the class will be randomly assigned either to get the teacher-AI or the answerer-AI, five students assigned to each.

24. Does this change the design of the study? If so, what is the new design and how have the components of the experiment changed (response, factors, units, etc)?

25. Propose a mechanism by which interference could occur in this study. Describe how you could redesign the study to prevent it.

26. From whose perspective is interference a more serious concern: a teacher or a cognitive scientist? Justify your answer.

27. Propose a mechanism by which noncompliance could occur in this study. Describe how you could redesign the study to prevent it.
28. The researchers are considering administering a short six question pre-test before the lesson to assess students' understanding of basic ideas in statistics. Describe *two* different ways in which you could use this data to improve your analysis (note there are at least three ways).
29. *This question is ungraded but I am curious about your thoughts:* This is a real study being designed by two Berkeley students and advised by Professors Grossman and Bray. It will be piloted this summer and conducted in the fall. Do you have any ideas or feedback that could help improve this study?

Scratch Paper

Any work here will be ungraded.